Poker Exercises 2

- A poker hand contains 5 out of 52 cards. The cards are 13 kinds (1-Ace, 2,..,10,J,Q,K) and 4 suites (spades, hearts, clubs, diamonds)
- How many hands are royal flush? Royal flush is {10,J,Q,K,Ace} of the same suit. This is the strongest card in poker.
- There are 4 royal flushes/one per suit
- How many hands are straight flush? A straight flush is a hand with all five cards of the same suit in sequential order starting at 1,2,….,9. 10 is excluded because it would lead to a Royal flush.
- I have C(9,1)=9 options for the beginning of straight flush, and 4 options for the suit, yielding a total of $4 \cdot 9=36$. There is a single option for each of the remaining cards.
- How many hands are straight? A straight is a hand with all five cards in sequential order. At least two cards are of different suit; otherwise, it would be a royal or straight flush.
- I will first count all straights including straight and royal flushes.
- I have C(10,1)=10 options for the beginning of the straight. For each of the five cards, there are 4 suite options. Therefore, total number of straights is: $10 \cdot 4^5$.
- If I exclude the straight and royal flushes, the number becomes: $10 \cdot 4^5 - 40$.